Software Requirements Specification

for

NTCentinel

Version 1.0 approved

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1. Introduction

1.1 Purpose

This documentation provides a detailed description of the software and hardware requirements for NTCentinel, a kiosk-based system built to track student violations at National Teachers College. The main purpose of this system is to transition away from the old method of pen and paper at the gate and replace it with an automated system. By using a system, it is much more efficient for guards to record offenses, ensure the data is accurate for the Office of Student Affairs (OSA), and help keep the gate area clear so students aren't crowded on the street.

1.2 Product Scope

The NTCentinel system is designed to modernize the way National Teachers College handles student violations at the main entrance. The project scope focuses on replacing the current process with a digital system. By automating the encoding and verification process, the system aims to further boost the flow of students into the campus. This is intended to solve the fundamental issue of student congestion on the sidewalk, in doing so, moves students out of potential risks and improves overall street safety.

As the main feature suggests, the system will handle all the previous manual processes digitally, from the scanning of IDs, checking the database for previous offenses, and instantly logging new violations. For the Office of Student Affairs (OSA), the system also includes a secure backend that ensures all data is accurate and protected from unwanted data access or leaks. Furthermore, the system will implement a feature to track how many offenses a student has accumulated, automatically flagging those who have reached the limit for official sanctions.

1.3 References

- Republic Act No. 10173 (Data Privacy Act of 2012).

2. Overall Description

2.1 Product Perspective

NTCentinel is a new system that combines custom hardware with a centralized digital network. Rather than being just a simple mobile app or a basic website. This means it uses physical hardware which is the kiosk terminal to interact with the users at the school gate, and then communicates instantly with the OSA central database over the school's network.

The system acts as the main bridge between the security guards and the administration. Before NTCentinel, the gate and the Office of Student Affairs were disconnected, relying on paper logs that had to be moved and encoded manually. This setup ensures that the information the guard sees at the gate is always synced with the records kept by the OSA, making the entire process connected.

2.2 Product Functions

- **Student Identity Verification:** Scanning ID QR codes to pull student data.
- **Violation Logging:** Digital selection of offense types by the Guard.
- **Automatic Counting:** Real-time tracking of offense frequency per student.
- **Slip Generation:** Printing physical violation receipts for students.
- **OSA Dashboard:** Centralized view for staff to monitor and manage records.

2.3 User Classes and Characteristics

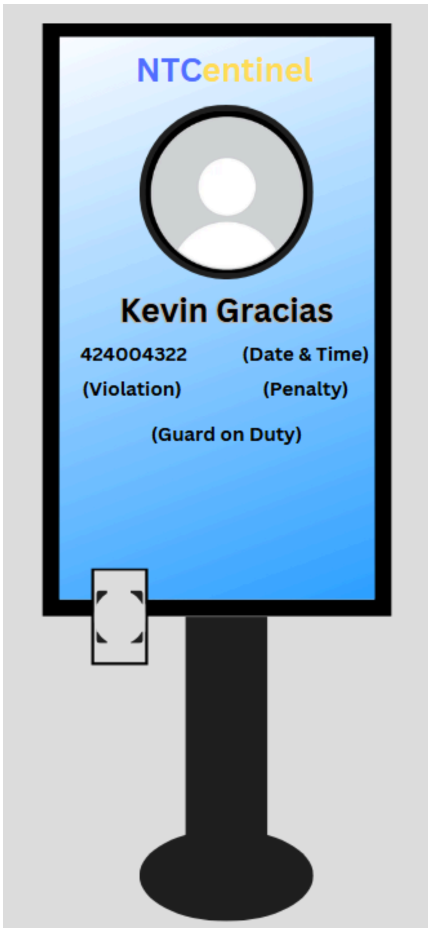
- **Security Guard:** Minimal technical expertise; requires a simple touchscreen interface for fast data entry.
- **OSA Staff:** Moderate technical expertise; requires data management and reporting capabilities.
- **Students:** Interact with the kiosk for identity verification.

2.4 Operating Environment

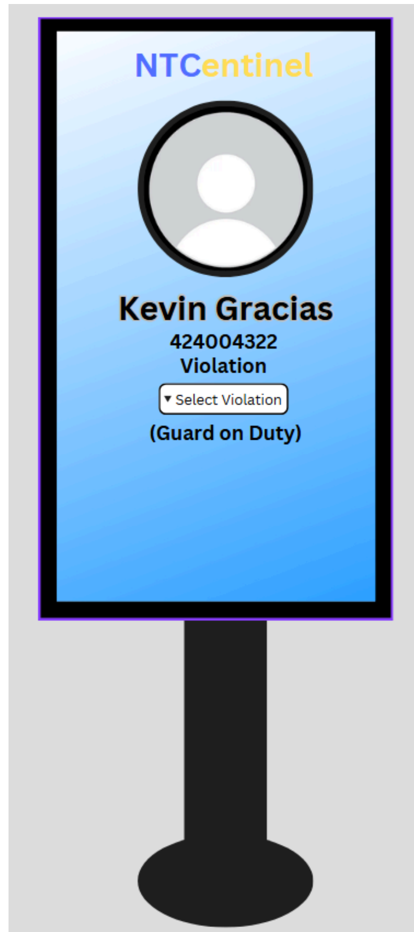
The system is placed nearby the entrance, running on a simplified application. It requires constant connection to the school's Local Area Network (LAN) to sync with the OSA server.

3. External Interface Requirements

3.1 User Interfaces



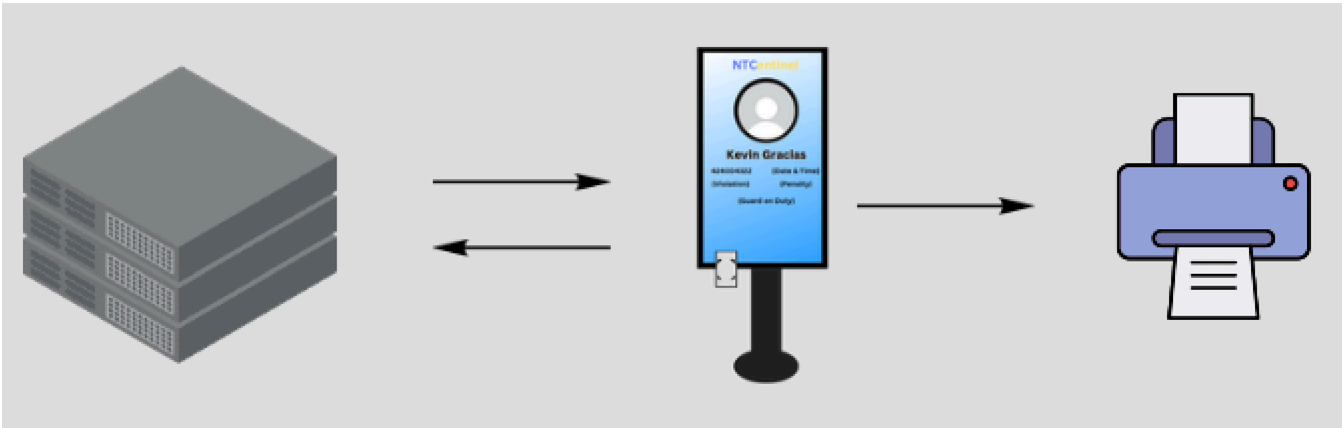
(Students UI)



(Guards UI)

The NTCentinel kiosk will use a large touchscreen. Since the kiosk will be placed at the school gate, the screen needs to be bright enough to see clearly even when it's sunny outside. The actual design of the app will focus on simplicity and user-friendly. The UI will implement big buttons and high-contrast colors so the guards operate through the menus without making mistakes.

3.2 Hardware Interfaces



For the kiosk to work, the software needs to talk to three main pieces of hardware. First is a QR Scanner that can quickly read a student’s ID. Second is a Small Thermal Printer to print out a violation slip for the student. Finally, it needs a Network connection so it can send all the data back to the OSA office in real-time.

3.3 Software Interfaces

NTCentinel isn't just a standalone box; it connects to the school's existing digital systems. It will link up with the NTC Student Database to automatically pull up names and photos when an ID is scanned. It also connects to a Central Database to save every violation and track how many offenses a student has. Lastly, we’ll set it up to use a Messaging/Email service so it can automatically alert the OSA staff the moment someone hits their 3rd violation.

3.4 Communications Interfaces

To keep everything safe, the kiosk will send all student data using local connection. This is so student info won't be exposed from outside connection while it's traveling from the gate to the server. If the school system goes down for a minute, the kiosk is designed to save the data temporarily first and then upload it automatically as soon as the connection comes back.

4. System Features

4.1 Student Identity Verification

4.1.1 Description and Priority

This feature serves as the system's primary entry point by capturing student identification data through an integrated ID QR scanner to verify active enrollment. Given that all subsequent logging depends on accurate identification, this feature is designated as High Priority. It provides a high relative benefit by eliminating the need for manual data entry, while maintaining a low implementation risk due to the maturity of QR scanning technology.

4.1.2 Response Sequences

The process begins when a student taps their ID on the kiosk's scanner. Once detected, the system activates the sensor to decode the QR printed behind the ID. When the data is captured, the system proceeds to query the Student records, which then responds by displaying the student's profile picture, full name, and current violation count on the kiosk's touchscreen for the guard to verify.

4.1.3 Functional Requirements

- REQ-01: The system shall successfully decode valid QR code within a maximum speed of 1 second upon detection.

4.2 Digital Violation Logging

4.2.1 Description and Priority

This feature provides the Security Guard with a digitized touchscreen interface to record specific academic offenses. It is a High Priority feature that directly addresses the "messy handwriting" issue identified in Phase 1. It carries a high benefit rating for data clarity.

4.2.2 Response Sequences

When a guard selects a primary violation category such as "Uniform" on the touchscreen, the system responds by populating a sub-menu of specific offenses like "No ID" or "Incomplete Uniform." After the guard selects the relevant offense and presses "Confirm," the system responds by updating the record in the central database.

4.2.3 Functional Requirements

- REQ-03: The system shall provide a digitized interface for guards to select violation types from a preset list.

- REQ-04: Access to the violation database and reports shall be restricted to authorized OSA personnel via role-based login and encrypted passwords.

4.3 Student Identity Verification

4.1.1 Description and Priority

This feature handles the "3-Strike Rule" logic by ensuring that every violation logged at the gate is immediately reflected in the school's central records. It is ranked as Medium Priority because it ensures that the student's history is always up to date for the Office of Student Affairs.

4.1.2 Response Sequences

Upon the guard's confirmation of a logged violation, the system stimulates a calculation sequence that increments the student's total offense count. The system then automatically syncs this data with the OSA database so that any necessary sanctions can be flagged.

4.1.3 Functional Requirements

- REQ-05: The system shall automatically update the OSA database without manual encoding.

4.4 Student Identity Verification

4.1.1 Description and Priority

This feature acts as a real-time bridge between the school gate and the administrative offices by flagging critical offenders. It is a High Priority feature with a benefit rating of 9, as it allows the OSA to act immediately on high-frequency violators without waiting for manual reports.

4.1.2 Response Sequences

When a student's violation count reaches the critical threshold of the 3rd offense, the system triggers a flag within the database. This alert is immediately visible on the OSA's monitoring dashboard, allowing for instant intervention..

4.1.3 Functional Requirements

- REQ-06: The system shall automatically flag students who reach three violations and notify the OSA staff.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

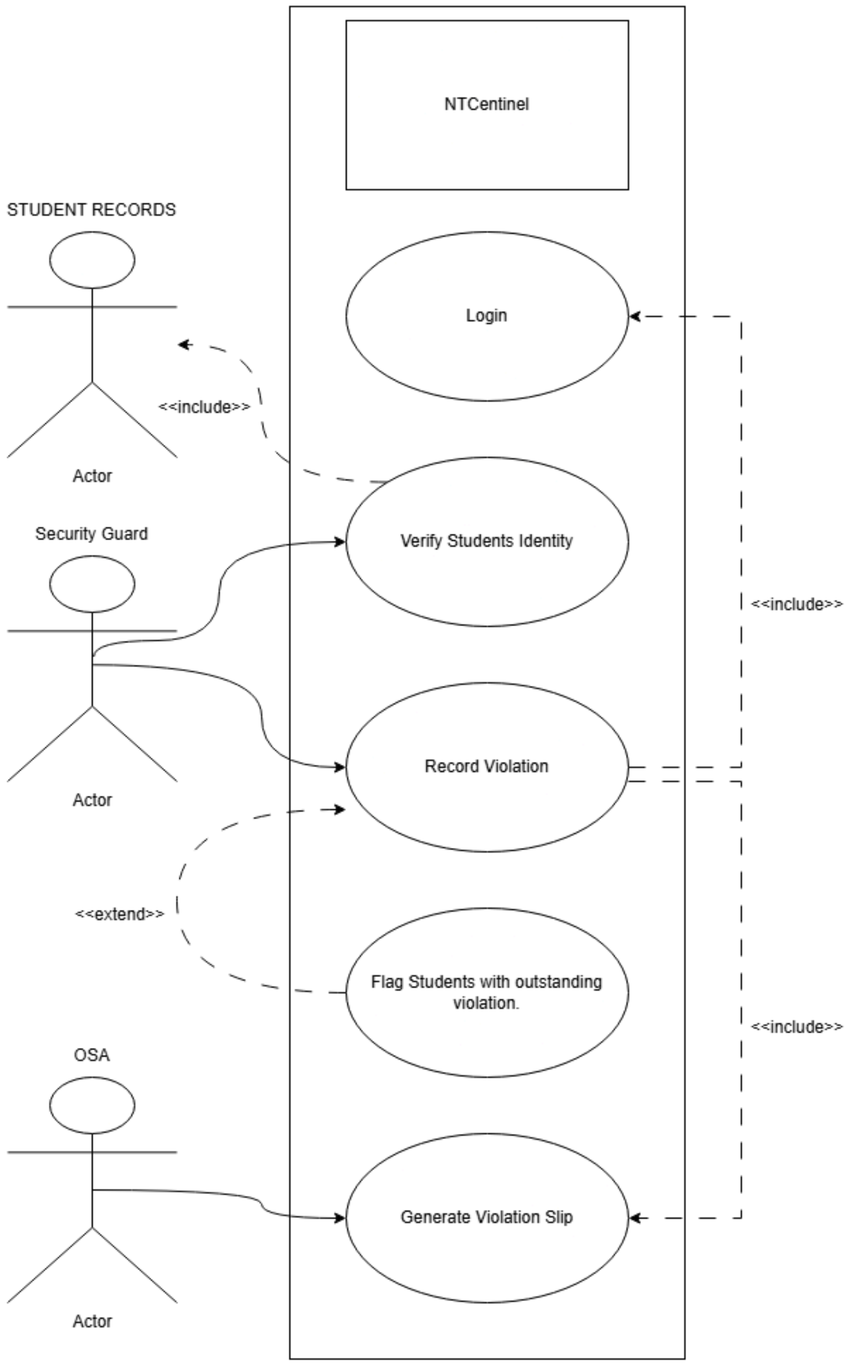
The priority goal for NTCentinel is efficiency. To relieve the congested gate, the system has to finish the entire process from scanning the ID to saving the record in under 3 seconds (REQ-02). If it takes any longer, students will start crowding the sidewalk and the street, which creates an unnecessary hazard.

5.2 Security Requirements

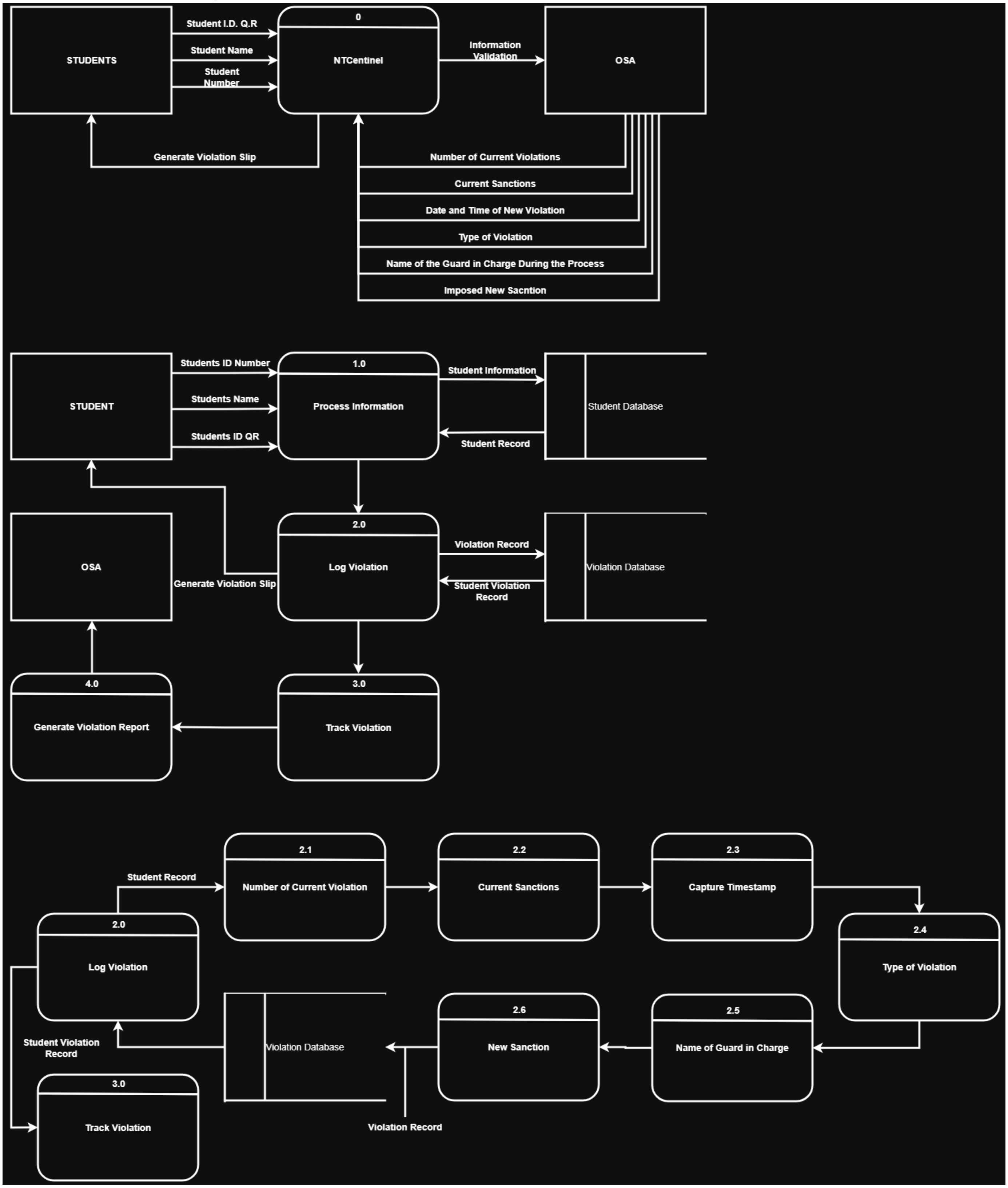
Since we are dealing with private student records, security is a must. Only the authorized personnel will be able to log in to the system access or modify the records (REQ-04). The system shall restrict data access to OSA Staff and authorized personnel via role-based login

Appendix A: Analysis Models

- **Use Case Diagram**



- **Data Flow Diagram**



- **Entity Relationship Diagram**